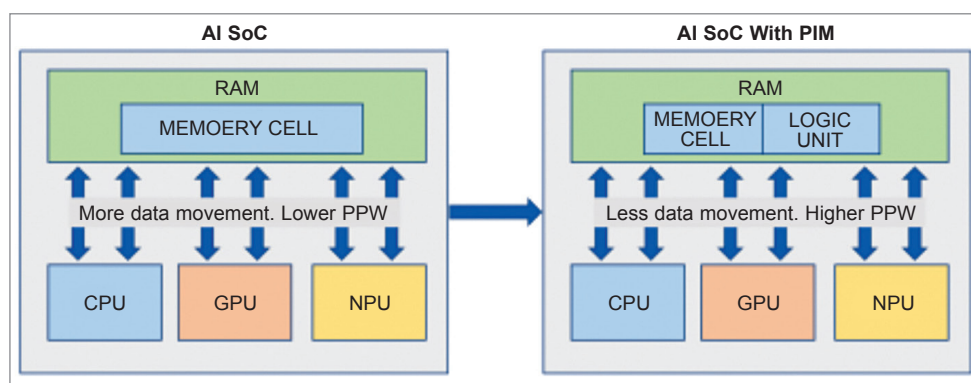


The Case Of PROCESSING-IN-MEMORY For AI SoCs

Processing-in-memory (PIM) has the potential to revolutionise AI SoCs, promising game-changing enhancements in speed, efficiency, and sustainability



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Emerging artificial intelligence (AI) applications will be more compute and memory-intensive than ever. The system-on-chip (SoC) architecture is crucial to manage such applications efficiently.

These SoCs rely on data pipelines and memory organisations to process the request. However, with every new AI application, the amount of data that needs to be processed is increasing 100 times, and the SoC design innovation cannot match it. Eventually, traditional SoCs will not be able to cater to the heavy demands of AI applications. Thus, several AI companies have started focusing on customised SoC solutions.

The emergence of AI SoC

This is where AI SoC also comes into the picture—Artificial intelligence system-on-a-chip. AI SoC uses either the aggregated (single silicon die) or disag-

gregated (multiple die, i.e., Chiplets) design and manufacturing methodology. What sets these apart from the traditional SoCs is that they are custom-designed to provide various types of processing elements (PEs)—CPU, GPU, NPU, etc, to ensure data is processed by maximising the number of tasks per given cycle.

Common challenges in AI SoCs

However, AI SoCs still suffer from the same issues as traditional SoCs. The data has to be moved continuously from lower-level memories (DRAM/SRAM) to upper-level memories (L3/L2/L1/registers). AI data-driven applications are in continuous contention to process as much data as possible, making AI SoCs spend clock cycles to move data from upper to lower-level memories and vice versa. All of this leads to slow processing and increases power consumption.